

PSTAT 126: REGRESSION ANALYSIS

Homework 1

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Question 1: The value of the parameter obtained in simple linear regression model with no intercept i.e., $y_i = \beta_1 x_i + \epsilon_i$ ($i = 1, 2, \dots, n$) using least square method is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2}$$

1. Is $\hat{\beta}_1$ biased or unbiased? Prove.

2. Find the variance of $\hat{\beta}_1$.

$$\begin{aligned} \text{Q. } E[\hat{\beta}_1] &= E\left[\frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2}\right] = \left[\frac{1}{\sum_{i=1}^n x_i^2}\right] \cdot E\left[\sum_{i=1}^n x_i y_i\right] = \sum_{i=1}^n \frac{1}{\sum_{i=1}^n x_i^2} \cdot \sum_{i=1}^n x_i E[y_i] \\ &= \sum_{i=1}^n \frac{1}{\sum_{i=1}^n x_i^2} \cdot \sum_{i=1}^n x_i E[\beta_1 x_i + \epsilon_i] = \sum_{i=1}^n \frac{1}{\sum_{i=1}^n x_i^2} \cdot \sum_{i=1}^n x_i [\beta_1 x_i] = \\ &= \frac{1}{\sum_{i=1}^n x_i^2} \cdot \sum_{i=1}^n \beta_1 x_i^2 = \frac{\beta_1 \sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2} = \beta_1. \end{aligned}$$

$$E[\hat{\beta}_1] = \beta_1 \Rightarrow \hat{\beta}_1 \text{ is unbiased.}$$

$$\text{Q. } \text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2}\right) = \frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2} \text{Var}\left(\sum_{i=1}^n y_i x_i\right)$$

$$\begin{aligned} &= \frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2} \cdot \sum_{i=1}^n x_i^2 \cdot \text{Var}(y_i) + 2 \sum_{i \neq j} \sum_{j=1}^n x_i x_j \text{Cov}(y_i, y_j) \\ &= \frac{1}{\left(\sum_{i=1}^n x_i^2\right)^2} \cdot \sum_{i=1}^n x_i^2 \cdot \sigma^2 \end{aligned}$$

since all y_i 's are independent,
 $\text{Cov}(y_i, y_j), i \neq j = 0$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n x_i^2}$$

Question 2: Simplify each of the following summations and verify the identities.

$$1. \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i = \sum_{i=1}^n (y_i - \bar{y})x_i = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

$$2. \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$3. \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$

$$\begin{aligned} 1. \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) &= \sum_{i=1}^n x_i y_i - x_i \bar{y} - y_i \bar{x} + \bar{x} \bar{y} \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y} - \sum_{i=1}^n y_i \bar{x} + \sum_{i=1}^n \bar{x} \bar{y} = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} \\ &= \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} \checkmark \end{aligned}$$

$$\sum_{i=1}^n (x_i - \bar{x})y_i = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n y_i \bar{x} = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} \checkmark$$

$$\sum_{i=1}^n (y_i - \bar{y})x_i = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y} = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} \checkmark$$

$$\begin{aligned} 2. \sum_{i=1}^n (x_i - \bar{x})^2 &= \sum_{i=1}^n x_i^2 - 2x_i \bar{x} + \bar{x}^2 = \sum_{i=1}^n x_i^2 - \sum_{i=1}^n 2x_i \bar{x} + \sum_{i=1}^n \bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 \checkmark \end{aligned}$$

$$\begin{aligned} 3. \sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n y_i^2 - \sum_{i=1}^n 2y_i \bar{y} + \sum_{i=1}^n \bar{y}^2 \\ &= \sum_{i=1}^n y_i^2 - 2n\bar{y}^2 + n\bar{y}^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2 \checkmark \end{aligned}$$

Question 3: Let Y be a random variable and X be a fixed variable. Sometimes it is useful to measure the independent variable around its mean. In such a case, the model has a centered version:

$$y_i = \beta_0 + \beta_1(x_i - \bar{x}) + \beta_1\bar{x} + \epsilon_i \quad (i = 1, 2, \dots, n)$$

$$= \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i$$

where $\beta_0^* = \beta_0 + \beta_1\bar{x}$.

1. Find the value of the parameters $\hat{\beta}_0^*$ and $\hat{\beta}_1$.
2. Are these parameters biased or unbiased? Prove.
3. Find the variance of the estimated value of parameters $\hat{\beta}_0^*$ and $\hat{\beta}_1$.

$$1. \min \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0^* - \beta_1(x_i - \bar{x}))^2 = Y \quad \sum_{i=1}^n (x_i - \bar{x}) = 0$$

$$\frac{\partial Y}{\partial \beta_0^*} = \sum_{i=1}^n -2(y_i - \beta_0^* - \beta_1(x_i - \bar{x})) = 0$$

$$\sum_{i=1}^n y_i - \sum_{i=1}^n \beta_0^* - \sum_{i=1}^n \beta_1(x_i - \bar{x}) = 0$$

$$n\bar{y} - n\beta_0^* - \beta_1 \sum_{i=1}^n (x_i - \bar{x}) = 0$$

$$n\bar{y} - n\beta_0^* = 0$$

$$\boxed{\hat{\beta}_0^* = \bar{y}}$$

$$\frac{\partial Y}{\partial \beta_1} = \sum_{i=1}^n -2(y_i - \beta_0^* - \beta_1(x_i - \bar{x}))(x_i - \bar{x}) = 0$$

$$= \sum_{i=1}^n (x_i - \bar{x})y_i - \sum_{i=1}^n (x_i - \bar{x})\beta_0^* - \sum_{i=1}^n \beta_1(x_i - \bar{x})^2 = 0$$

$$= \sum_{i=1}^n S_{xy} - \beta_0^* \sum_{i=1}^n (x_i - \bar{x}) - \beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 = 0$$

$$= S_{xy} - \beta_1 S_{xx} = 0$$

$$= S_{xy} = \beta_1 S_{xx}$$

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

$$2. E[\hat{\beta}_1] = E\left[\frac{S_{xy}}{S_{xx}}\right] = E\left[\frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\right] = \frac{\sum_{i=1}^n (x_i - \bar{x}) E[y_i]}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{1}{S_{xx}} \cdot \sum_{i=1}^n (x_i - \bar{x}) E[\beta_0 + \beta_1(x_i - \bar{x}) + \beta_1\bar{x} + \epsilon_i] = \frac{1}{S_{xx}} \cdot \sum_{i=1}^n (x_i - \bar{x}) [\beta_0 + \beta_1(x_i - \bar{x}) + \beta_1\bar{x}]$$

$$= \frac{1}{S_{xx}} \cdot \left[\sum_{i=1}^n \beta_0(x_i - \bar{x}) + \sum_{i=1}^n \beta_1(x_i - \bar{x})^2 + \sum_{i=1}^n \beta_1(x_i - \bar{x})\bar{x} \right]$$

$$= \frac{1}{S_{xx}} \left[\beta_1 \cdot \sum_{i=1}^n (x_i - \bar{x})^2 + \beta_1 \cdot \left[\sum_{i=1}^n (x_i - \bar{x})\bar{x} \right] \right] = \frac{1}{S_{xx}} \left[\beta_1 \cdot S_{xx} + \beta_1(n\bar{x}^2 - n\bar{x}^2) \right]$$

$$= \frac{1}{S_{xx}} \cdot (\beta_1 \cdot S_{xx}) = \beta_1 \quad \checkmark$$

[backside]

$\hat{\beta}_1$ is unbiased 4/7

3)

$$2. E[\hat{\beta}_0^*] = E[\bar{y}] = E\left[\frac{1}{n} \sum_{i=1}^n y_i\right] = \frac{1}{n} E\left[\sum_{i=1}^n \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i\right]$$

$$= \frac{1}{n} \left[E\left[\sum_{i=1}^n \beta_0^*\right] + E\left[\sum_{i=1}^n \beta_1(x_i - \bar{x})\right] + E\left[\sum_{i=1}^n \epsilon_i\right] \right]$$

$$= \frac{1}{n} \sum_{i=1}^n E[\beta_0^*] = \frac{1}{n} \cdot n \beta_0^* = \beta_0^* \checkmark$$

$\Rightarrow \hat{\beta}_0^*$ is unbiased

$$3. \text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\right) = \frac{1}{\sum_{i=1}^n x_i^2} \cdot \text{Var}\left(\sum_{i=1}^n (x_i - \bar{x}) y_i\right)$$

$$= \frac{1}{\sum_{i=1}^n x_i^2} \cdot \left[\sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(y_i) + 2 \sum_{i < j} (x_i - \bar{x})(x_j - \bar{x}) \text{Cov}(y_i, y_j) \right]$$

$$= \frac{1}{\sum_{i=1}^n x_i^2} \cdot \sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n x_i^2} \cdot \sigma^2$$

$$= \boxed{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}$$

$$\begin{aligned} \text{Var}(\hat{\beta}_0^*) &= \text{Var}(\bar{y}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n y_i\right) = \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(y_i) + 2 \sum_{i < j} \text{Cov}(y_i, y_j) \right] \\ &= \frac{1}{n^2} \left[\sum_{i=1}^n \sigma^2 \right] = \frac{n \sigma^2}{n^2} = \boxed{\frac{\sigma^2}{n}} \end{aligned}$$

Question 4: Prove that an estimator of β_0 , i.e., $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ is a linear combination of y_i .

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{let } k_i = \frac{x_i - \bar{x}}{S_{xx}}$$

$$= \sum_{i=1}^n k_i y_i \Rightarrow \hat{\beta}_1 \text{ is a linear combination of } y_i$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n \frac{1}{n} y_i - \sum_{i=1}^n k_i y_i \bar{x}$$

$$= \sum_{i=1}^n \left[\frac{1}{n} - k_i \bar{x} \right] y_i = \sum_{i=1}^n c_i y_i$$

$$\text{let } c_i = \frac{1}{n} - k_i \bar{x} \Rightarrow \hat{\beta}_0 \text{ is a linear combination of } y_i$$

Question 5: Prove that $\hat{\sigma}^2$ obtained using least squares method in simple linear regression model is unbiased.

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n e_i^2}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$$

$$\begin{aligned} \sum_{i=1}^n (y_i - \hat{y}_i)^2 &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 (x_i - \bar{x}))^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (\hat{\beta}_1 (x_i - \bar{x}) + (e_i - \bar{e}))^2 = \sum_{i=1}^n [(e_i - \bar{e}) - (\hat{\beta}_1 - \beta_1)(x_i - \bar{x})]^2 \end{aligned}$$

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{e}$$

$$y_i - \bar{y} = \beta_1 (x_i - \bar{x}) + (e_i - \bar{e})$$

$$\begin{aligned} e_i &= y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \\ &= y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i \\ &= (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \end{aligned}$$

$$\begin{aligned} e_i &= [\beta_1 (x_i - \bar{x}) + (e_i - \bar{e})] - \hat{\beta}_1 (x_i - \bar{x}) \\ &= (\beta_1 - \hat{\beta}_1)(x_i - \bar{x}) + (e_i - \bar{e}) \\ &= (e_i - \bar{e}) - (\hat{\beta}_1 - \beta_1)(x_i - \bar{x}) \end{aligned}$$

$$\sum_{i=1}^n (e_i - \bar{e})^2 = \sum_{i=1}^n e_i^2 - n\bar{e}^2 = \sum_{i=1}^n e_i^2 - \frac{(\sum e_i)^2}{n}$$

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x})e_i}{S_{xx}}$$

$$\sum_{i=1}^n (e_i - \bar{e})(x_i - \bar{x}) = \sum_{i=1}^n e_i (x_i - \bar{x}) = (\hat{\beta}_1 - \beta_1) S_{xx}$$

$$\Rightarrow 2(\hat{\beta}_1 - \beta_1) \sum_{i=1}^n (e_i - \bar{e})(x_i - \bar{x}) = 2(\hat{\beta}_1 - \beta_1)^2 S_{xx}$$

$$\sum_{i=1}^n E[e_i^2] = \sum_{i=1}^n \text{Var}(e_i) + [E(e_i)]^2 = n\sigma^2$$

$$\begin{aligned} E\left[\sum_{i=1}^n (x_i - \bar{x})e_i\right] &= \sum_{i=1}^n E[(x_i - \bar{x})e_i] \\ &= \sum_{i=1}^n (x_i - \bar{x}) \text{Cov}(e_i) - E[(x_i - \bar{x})e_i]^2 \\ &= S_{xx} \sigma^2 \end{aligned}$$

$$= \sum_{i=1}^n [(e_i - \bar{e})^2 - 2(\hat{\beta}_1 - \beta_1)(e_i - \bar{e})(x_i - \bar{x}) + (\hat{\beta}_1 - \beta_1)^2 (x_i - \bar{x})^2]$$

$$= \sum_{i=1}^n (e_i - \bar{e})^2 - 2(\hat{\beta}_1 - \beta_1) \sum_{i=1}^n (e_i - \bar{e})(x_i - \bar{x}) + (\hat{\beta}_1 - \beta_1)^2 \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \sum_{i=1}^n e_i^2 - \frac{(\sum_{i=1}^n e_i)^2}{n} - 2(\hat{\beta}_1 - \beta_1) S_{xx} + (\hat{\beta}_1 - \beta_1)^2 S_{xx}$$

$$= \sum_{i=1}^n e_i^2 - \frac{(\sum_{i=1}^n e_i)^2}{n} - (\hat{\beta}_1 - \beta_1)^2 S_{xx}$$

$$= \sum_{i=1}^n e_i^2 - \frac{(\sum_{i=1}^n e_i)^2}{n} - \frac{(\sum_{i=1}^n (x_i - \bar{x})e_i)^2}{S_{xx}} = SSE$$

$$\begin{aligned} E[SSE] &= E\left[\sum_{i=1}^n e_i^2\right] - E\left[\frac{(\sum_{i=1}^n e_i)^2}{n}\right] - E\left[\frac{(\sum_{i=1}^n (x_i - \bar{x})e_i)^2}{S_{xx}}\right] \\ &= n\sigma^2 - \sigma^2 - \sigma^2 = (n-2)\sigma^2 \end{aligned}$$

$$E[\hat{\sigma}^2] = E\left[\frac{SSE}{n-2}\right] = \frac{1}{n-2} E[SSE]$$

$$= \frac{1}{n-2} \cdot (n-2)\sigma^2 = \sigma^2 \checkmark$$

$\Rightarrow \therefore \hat{\sigma}^2$ is unbiased

Question 6: You are an environmental scientist. As we all know, higher temperatures in summer can lead to increased usage of air conditioning, driving higher energy consumption. One day, you decide to create a model to forecast energy demand. After collecting 50 observations, you analyze the dataset provided.

SCENARIO -1

Initially, you realize that you forgot to observe temperature and only have data on energy consumption.

TASKS

1. Create a null model using the energy consumption data and write the intercept for the model.
2. Interpret the intercept of the null model. What does it represent in the context of this problem?
3. Evaluate the null model's goodness of fit using R^2 .

SCENARIO -2

With the help of a colleague, you manage to retrieve the temperature data. Now, you want to explore the relationship between temperature and energy consumption by creating a simple regression model.

TASKS

1. Create a scatterplot to visualize the relationship between temperature and energy consumption.
2. Fit a simple linear regression model to the data and provide the summary of the model.
3. Interpret the coefficients of the regression model. What do the intercept and slope represent in the context of this problem?
4. Evaluate the model's goodness of fit using R^2 .

FINAL QUESTION

After completing both models, which model (null or simple regression) would you prefer for further analysis, and why?

6. Scenario 1

2. The intercept is 102.856. This means the average energy consumed is 102.856 kWh, which will be the prediction outcome for any test prediction.

3. R^2 for a null model will always be 0 because there are no other variables to measure any thing against ($SSE = SST$)

Scenario 2

3. $\beta_0^* = 50.93$ (the intercept). This means that when the temperature is 0°C , the predicted energy consumption is approximately 50.93 kWh.

$\beta_1^* = 2.455$ (slope). This means that for every 1°C increase in temperature, the energy consumption is predicted to increase by 2.455 kWh.

4. $R^2 = 0.9374$ meaning about 93.74% of the variance in energy use is explained by temperature.

Final Question

I would use the simple regression model because a lot more of the variance in energy consumption is captured by this model than the null model, making it better for future analysis.

hw1

Mitul Marimuthu

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```
data <- read.csv("./datasets/Dataset.csv")
```

```
class(data['Energy_Consumption_kWh'])  
head(data['Energy_Consumption_kWh'])
```

```
null_model <- lm(Energy_Consumption_kWh ~ 1, data = data)
```

```
null_model$coefficients
```

```
## (Intercept)  
##      102.8555
```

```
mean(data$Energy_Consumption_kWh)
```

```
## [1] 102.8555
```

```
sse_null <- sum(resid(null_model)^2)  
sse_null
```

```
## [1] 16436.42
```

```
summary(null_model)
```

```
##  
## Call:  
## lm(formula = Energy_Consumption_kWh ~ 1, data = data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -28.178 -14.780   0.021  11.335  39.273   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   102.86         2.59   39.71  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 18.31 on 49 degrees of freedom
```

```
sim_lin_reg = lm(data$Energy_Consumption_kWh ~ data$Temperature_C, data = data)
```

```
summary(sim_lin_reg)
```

```
##
## Call:
## lm(formula = data$Energy_Consumption_kWh ~ data$Temperature_C,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5034  -2.4541   0.0241   3.4235   8.3998
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    50.93031     2.04439   24.91  <2e-16 ***
## data$Temperature_C  2.45531     0.09158   26.81  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.63 on 48 degrees of freedom
## Multiple R-squared:  0.9374, Adjusted R-squared:  0.9361
## F-statistic: 718.8 on 1 and 48 DF,  p-value: < 2.2e-16
```

```
sse_lin <- sum(resid(sim_lin_reg)^2)
```

```
sse_null
```

```
## [1] 16436.42
```

```
sse_lin
```

```
## [1] 1028.839
```

The simple linear regression model has a smaller SSE, therefore it is a better model.

```
plot(data$Temperature_C, data$Energy_Consumption_kWh, xlab = "Temperature (C)",
      ylab = "Energy Consumption (kWh)", main = "Temperature vs Energy Consumption")
abline(null_model, col = "red")
abline(sim_lin_reg, col = "blue")
```

Temperature vs Energy Consumption

